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Employee Activity Dashboard - Oracle Database
Project Report

Assignment – ETL

Integrate multiple data sources like Timesheets, Event calendar and Absence Log, having different type and structure, model them into a Star schema and create queries/reports to show activities by day, by employees.

Solution:

1. **Project Overview**

This project implements an Oracle-based employee activity monitoring system. It integrates three different sources of information: internal timesheet data, absence records, and Outlook meeting data. The final result is a unified daily reporting structure exposed through a Streamlit web dashboard. The project flow described in the README is built around Script 1 through Script 8, with the final dashboard reading employees from EMPLOYEES and daily activity from VW_EMPLOYEE_DAY_BOARD.

2. **Data Sources**

a. Internal timesheet (Mock Data)

The internal timesheet data is demo data created to populate the application with realistic working activity for 15 employees. The population script creates weekly timesheets and inserts daily work entries with hours, projects, task descriptions, billing flags, and JSON metadata such as work mode and task type. The inserted dates show that the populated demo covers the weeks starting on **2026-03-23** and **2026-03-30**.

b. Absence Log Imported from Excel (Mock Data)

The absence source is an Excel-style raw dataset loaded into ABSENCES_RAW. It is intentionally irregular, with mixed cases, spacing issues, and inconsistent values, so that the sanitization process can demonstrate how raw business data is standardized and validated. The sanitization script reads ABSENCES_RAW, creates ABSENCES_SANITIZED, normalizes employee names and emails, standardizes absence codes, and validates business rules such as allowed hours and allowed weeks.

c. Meetings extracted from Outlook (Real Data)

The meeting source comes from Outlook. The Script 5.5 exports all the Outlook meeting participants into a CSV file, and the sanitization flow then loads and transforms this data into MEETINGS_SANITIZED. The meetings sanitization script cleans participant names and emails, removes room resources, removes cancelled meetings, eliminates duplicates.

3. Data Model and Reporting Structure

The project starts from a core operational Oracle schema built in Script 2. That schema contains the main tables:

- DEPARTMENTS
- EMPLOYEES
- PROJECTS
- TIMESHEETS
- TIMESHEET_ENTRIES

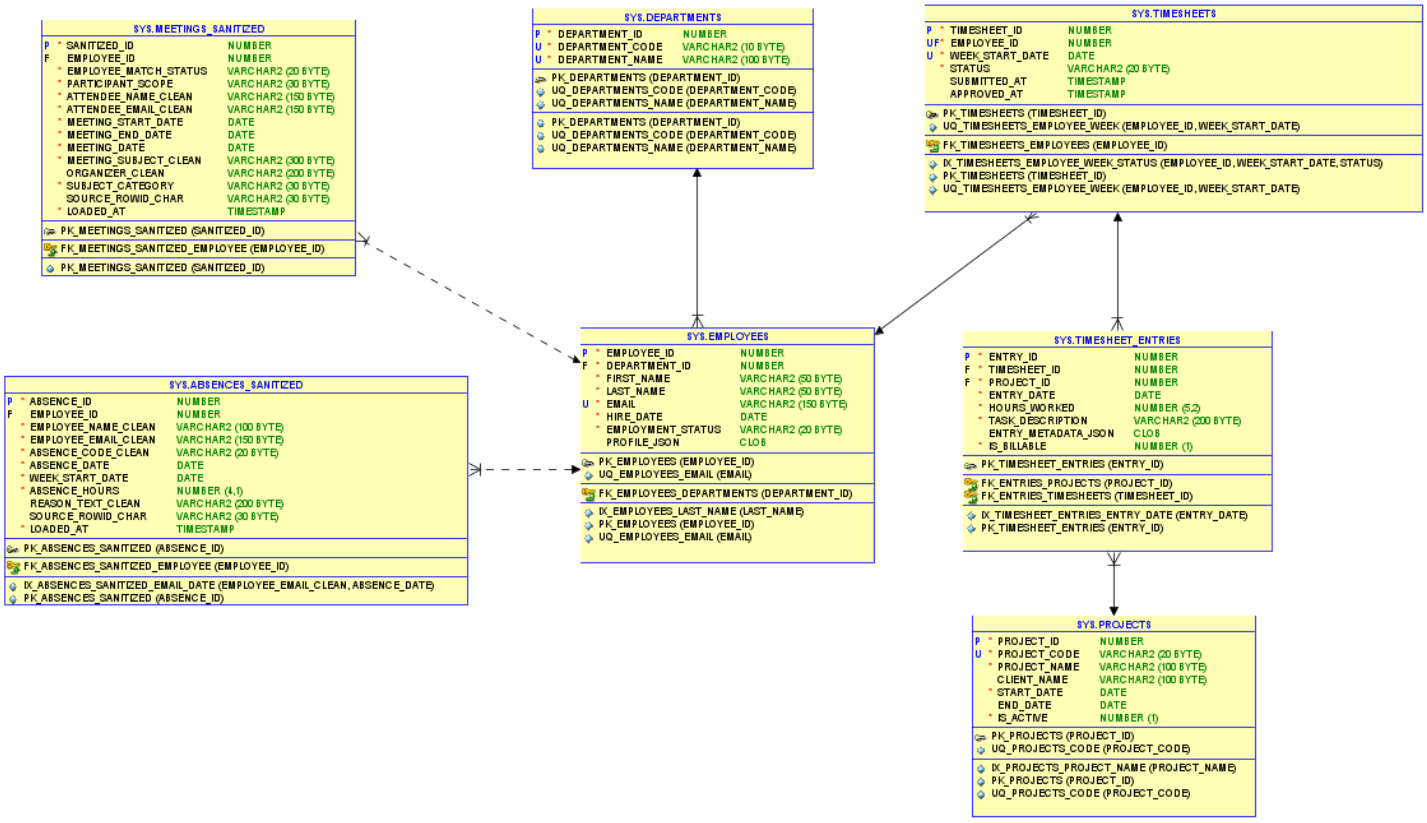
On top of this base model, the project adds:

- sanitized absence data in ABSENCES_SANITIZED
- sanitized meeting data in MEETINGS_SANITIZED
- reporting views that combine and summarize activity across sources.

From a reporting perspective, the final model works as a **star-schema-oriented structure**:

- the daily employee activity acts as the reporting fact layer
- employees, departments, projects, dates, absence types, and meeting categories act as descriptive dimensions
- the final integrated reporting view is VW_EMPLOYEE_DAY_BOARD

Star Schema / Data Model Diagram:



4. **How the Integration Works**
   The integration is implemented in stages:

   a. Script 1 two roles used by the HR to monitor the data.

   b. Script 2 creates the core Oracle database structure, including constraints, JSON columns, indexes, a normal view, a materialized view, audit logic, and security roles / grants. The original assignment report for the database layer explicitly lists constraints, JSON support, indexes, views, a materialized view, GROUP BY, LEFT JOIN, analytic functions, and audit logic as part of the implemented solution.

   c. Script 3 populates the internal timesheet data.

   d. Script 5 sanitizes the absence source into ABSENCES_SANITIZED and separates invalid rows into ABSENCES_REJECTED.

   e. Script 6 sanitizes the meeting source into MEETINGS_SANITIZED.

   f. Script 7 creates history and daily summary views that combine timesheet, absence, and meeting events. VW_EMPLOYEE_DAILY_SUMMARY aggregates worked hours, absence hours, and meeting count per employee and date.

   g. Script 8 creates VW_EMPLOYEE_DAY_BOARD, the final reporting view used by the dashboard. It aggregates work entries, absences, and meetings per employee and day.

5. **Running the Project**
   The execution order:
   - Script 1 - Role Creation
   - Script 2 - Timesheet System
   - Script 3 - Internal Timesheet Population
   - Script 4 - Absence Target / Staging
   - Script 5 - Absence Sanitization
   - Script 5.5 - Outlook Extraction
   - Script 6 - Meeting Sanitization
   - Script 7 - History and Daily Summary Views
   - Script 8 - Final Dashboard View
   - app.py - Streamlit web application.

5. **Web Application**

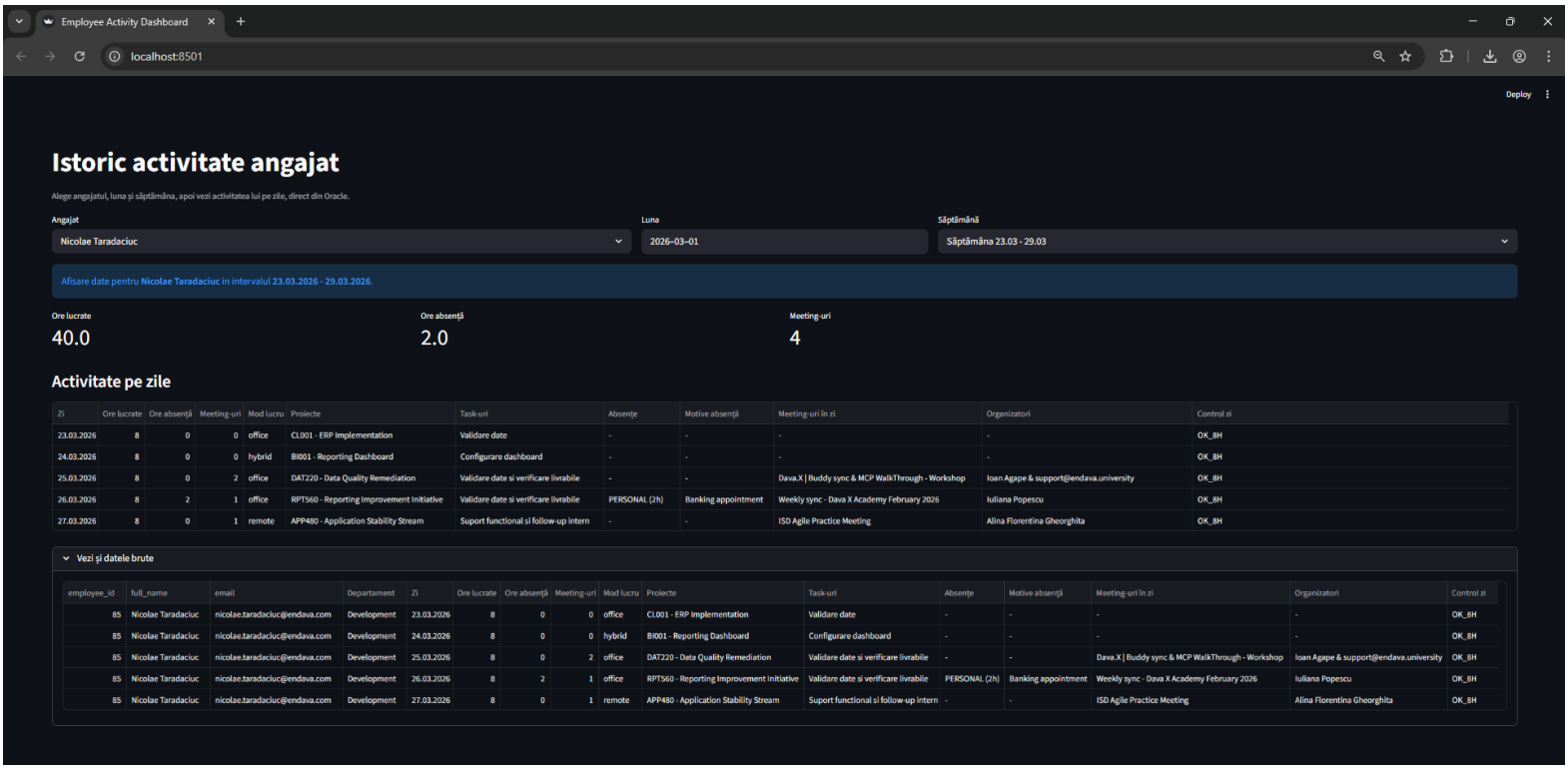
The web layer is implemented in Streamlit. The application connects to Oracle, reads employees from EMPLOYEES, and reads the final integrated activity data from VW_EMPLOYEE_DAY_BOARD.

It allows the user to select:
- an employee
- a month
- a week

Then it displays:

- worked hours
- absence hours
- meeting count
- a daily activity table including work mode, projects, tasks, absences, meeting list, organizers, and day control.

Dashboard Screenshot:



8. Final Conclusion

Employee Activity Dashboard is an Oracle-based ETL and reporting solution that integrates operational timesheet data, absence logs, and Outlook meeting data into one unified daily activity model. The database layer handles validation, constraints, sanitization, and reporting objects, while the Streamlit application provides a simple interface for viewing employee activity by day and by week.